Math-M-Addicts
MINDS THAT COUNT

# SP250 Homework 4

1. The rule for divisibility by 11 is as follows:

Consider any positive integer N. **Starting from the right**, compute the sum of all the digits in the even positions (call that sum E) and the sum of all the digits in the odd positions (call it O).

Then $N \equiv E - O \ (mod\ 11)$.

For example: N = 193257. E = 7 + 2 + 9 = 18. O = 5 + 3 + 1 = 9. According to the rule, N gives remainder 18 - 9 = 9 when divided by 11.

Prove this rule.

**Solution:**

Since $10 \equiv -1 \ (mod\ 11)$, we have $10^k \equiv (-1)^k \ (mod\ 11)$. So even powers of 10 are 1 (mod 11) and odd powers of 10 are -1 (mod 11).

Now, for a number $N$ with digits $a_k, a_{k-1}, ..., a_2, a_1, a_0$, where $a_k$ is the leftmost digit and $a_0$ is the rightmost digit, we have

$$N = 10^k a_k + 10^{k-1} a_{k-1} + ... + 10^2 a_2 + 10 a_1 + a_0$$
$$\equiv (-1)^k a_k + (-1)^{k-1} a_{k-1} + ... + (-1)^2 a_2 + (-1) a_1 + a_0 \ (mod\ 11)$$
$$\equiv E - O \ (mod\ 11)$$

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2. Prove, **using ONLY the material we have covered so far** and, in particular, no fancy factorizations, that for any two integers $a$ and $b$ with $a \neq b$ and positive integer $n$:

$$a - b \mid a^n - b^n.$$

**Solution:**

Since any nonzero integer divides itself, we have $(a - b) \mid (a - b)$. So, by definition of modular congruence,

$$a \equiv b \ mod\ (a - b)$$

Taking both sides of the above congruence to power n (we can do this because remainders multiply by the mod manipulation theorem we saw in class), we have

$$a^n \equiv b^n \ mod\ (a - b)$$

By the definition of modular congruence, this means $a - b \mid a^n - b^n$.

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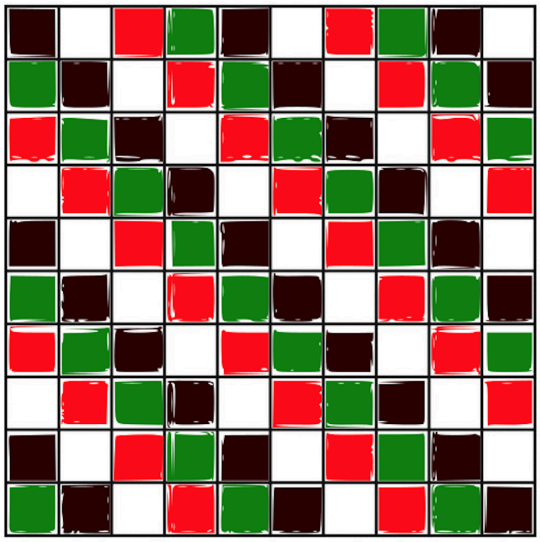
3. Prove that from any 100 integers one can pick 15 such that the difference between any two of them is divisible by 7.

**Solution:**
There are 7 possible remainders mod 7. We observe that $100 > 7 \cdot 14$. Hence, among any 100 integers there must be 15 that have the same remainder mod 7 by the pigeonhole principle. The difference between any two of the 15 integers is divisible by 7 since they all have the same remainder mod 7.

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4. Is it possible to tile a 10x10 square with 25 1x4 rectangles? For the avoidance of doubt, tiling means that every one of 100 little squares belongs to exactly one 1x4 rectangle.

**Solution:**
It is not possible. To see this, let's color the 10x10 board using 4 colors (black, white, red, and green) such that any 4 consecutive squares in any column or row consist of exactly one square of each color. One way to do this is shown below, where we colored the squares in the first row in the repeating sequence of (black, white, red, green), and shifted the repeating sequence by one square to the right for each subsequent row. In this coloring, there are 26 black squares, 25 white squares, 24 red squares, and 25 green squares. Since any 1x4 rectangle placed on the board will cover exactly one black square, there is no way for 25 1x4 rectangles to cover all 26 black squares. Therefore, it is not possible to tile a 10x10 board with 1x4 rectangles.



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5. Prove that equation $x^2 = 4y - 1$ does not have any integer solutions.

**Solution:**
Let's consider both sides of the equation in mod 4.

For the right hand side, we have $4y - 1 \equiv -1 \equiv 3 \ (mod\ 4)$.

For the left hand side, we see that squares can only be 0 or 1 mod 4

$$0^2 \equiv 0 \ \ (mod\ 4)$$
$$1^2 \equiv 1 \ \ (mod\ 4)$$
$$2^2 \equiv 0 \ \ (mod\ 4)$$
$$3^2 \equiv 1 \ \ (mod\ 4)$$

Since the left hand side of the equation can only be 0 or 1 mod 4 while the right hand side of the equation is always 3 mod 4, the equation cannot be satisfied by any choice of integers for x and y.

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6. a) What is the maximum number of chess rooks that one can place on a chessboard so that no two rooks attack each other?

b) What is the number of ways to place rooks in such a way?

**Solution:**

a) The maximum is 8. We can place no more than 8 rooks as there are 8 rows on the chessboard and at most one rook can be placed in each row. One way to place 8 rooks is to just put them all on one of the main diagonals of the chess board.

b) Let's place a rook in row one first, followed by placing a rook in row two, and so forth. For the first rook in row one, there are 8 possible positions to choose from. For the second rook in row two, there are only 7 possible positions to choose from as it cannot be placed in the same column as the first rook. Similarly, the third rook in row three will only have 6 possible positions to choose from, and so on. Hence, the total number of ways to place the rooks is

$$8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 40320$$

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